

SECTION-I
Straight Objective Type

This section contains **60 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- Which of the following are correct for sets A, B and C ?
 (1) $A - B = A - (A \cap B)$ (2) $A - B = A - (A \cup B)$
 (3) $A - (B \cup C) = (A - B) \cup (A - C)$ (4) $A - B = A \cup B'$
- In a certain town, 25% of the families own a phone and 15% own a car; 65% families own neither a phone nor a car and 2,000 families own both a car and a phone. Consider the following three statements :
 (i) 5% families own both a car and a phone (ii) 35% families own either a car or a phone
 (iii) 40,000 families live in the town
 Then,
 (1) Only (ii) and (iii) are correct (2) Only (i) and (iii) are correct
 (3) All (i), (ii) and (iii) are correct (4) Only (i) and (ii) are correct
- Let $A = \{(x, y) : x > 0, y > 0, x^2 + y^2 = 1\}$ and let $B = \{(x, y) : x > 0, y > 0, x^6 + y^6 \geq 1\}$. Then $A \cap B =$
 (1) A (2) B (3) ϕ (4) $\{(0, 1), (1, 0)\}$
- Let X be the set of all persons living in a state. Elements x and y in X are said to be related whenever y is 5 years older than X. The relation X is
 (1) reflexive only (2) symmetric only
 (3) symmetric & transitive (4) none of these
- Let $P = \{(x, y) : x^2 + y^2 = 1, x \in \mathbb{R}, y \in \mathbb{R}\}$ be a relation in R the set of real numbers then the relation P is
 (1) reflexive only (2) symmetric only
 (3) transitive only (4) reflexive and transitive only
- If $A = \{1, 2, 3, 4\}$ then the relation $R = \{(1, 3), (2, 4), (2, 3), (3, 1)\}$ defined on set A is
 (1) reflexive (2) symmetric (3) transitive (4) none of these
- Let P be the relation on the set of all real numbers such that $P = \{(a, b) : \sec^2 a - \tan^2 b = 1\}$. Then P is :
 (1) reflexive and symmetric but not transitive (2) reflexive and transitive but not symmetric
 (3) symmetric and transitive but not reflexive (4) an equivalence relation
- Let N_1 be the number of injective mappings from a set with m elements to a set with n elements when $m \leq n$. Now let number of injective mappings when $m > n$, is N_2 . Then
 (1) $N_1 > N_2$ (2) $N_1 < N_2$ (3) $N_1 = N_2$ (4) $N_1 + N_2 = m + n$

9. The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = (x - a)(x - b)(x - c)$ where $a, b, c \in \mathbb{R}$ is
 (1) not one-one but onto (2) one-one but not onto
 (3) both one-one and onto (4) neither one-one nor onto
10. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = \begin{cases} 2x + \alpha^2, & x \geq 2 \\ \frac{\alpha x}{2} + 10, & x < 2 \end{cases}$. If $f(x)$ is onto function then set of values of α is
 (1) $[1, 4]$ (2) $[-2, 3]$ (3) $(0, 3]$ (4) $[2, 5]$
11. If $f(x) = 1 + |x - 1|$, $-1 \leq x \leq 3$ and $g(x) = 2 - |x + 1|$, $-2 \leq x \leq 2$ then choose the appropriate option.
 (1) $\text{fog}(x) = x - 1$ for $x \in (0, 1)$ (2) $\text{fof}(x) = x$ for $x \in (-1, 1)$
 (3) $\text{gog}(x) = x$ for $x \in (-1, 2)$ (4) all of these
12. Domain of $y = \sqrt{\log_2 \left(\frac{x}{x+3} \right)}$ is
 (1) $(-\infty, -3)$ (2) $(-\infty, -3) \cup (0, \infty)$
 (3) $(0, \infty)$ (4) $(-\infty, \infty)$
13. Domain of $f(x) = \frac{1}{[\log_x 2]}$ where $[\cdot]$ is greatest integer function, contains the set
 (1) $(1, 3)$ (2) $[2, \infty)$ (3) $(0, 1)$ (4) $(1, 3]$
14. Let f be a function such that $f(x) = f(2-x)$ and $g(x) = f(x+1)$ then
 (1) $g(x)$ is an odd function
 (2) $g(x)$ an even function
 (3) graph of $g(x)$ is symmetrical about $x = 1$
 (4) graph of $f(x)$ is symmetrical about $(1, 0)$
15. The number of points, at which the two curves $y = \frac{x}{99}$ and $y = \sin \pi x$ intersect, is :
 (1) 199 (2) 198 (3) 200 (4) 201
16. Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a function such that $f(x) = x^2 - x \sin x - \cos x$. Then
 (1) $f(x)$ is one-one (2) $f(x)$ is onto
 (3) $f(x) = -2$ has one solution (4) $f(x)$ is many-one
17. The function $f(x) = |\sin 4x| + |\cos 2x|$, is a periodic function with fundamental period :
 (1) 2π (2) π (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{4}$
18. Let $A = \{x_1, x_2, \dots, x_7\}$ and $B = \{y_1, y_2, y_3\}$ be two sets containing seven and three distinct elements respectively. The total number of functions $f: A \rightarrow B$ that are onto, if there exist exactly three elements x in A such that $f(x) = y_2$, is equal to :
 (1) $14 \cdot {}^7C_3$ (2) $16 \cdot {}^7C_3$ (3) $14 \cdot {}^7C_2$ (4) $12 \cdot {}^7C_2$
19. For $x \in \mathbb{R}$, $x \neq 0$, $x \neq 1$, let $f_0(x) = \frac{1}{1-x}$ and $f_{n+1}(x) = f_0(f_n(x))$, $n = 0, 1, 2, \dots$. Then the value of $f_{100}(3) + f_1\left(\frac{2}{3}\right) + f_2\left(\frac{3}{2}\right)$ is equal to:
 (1) $\frac{4}{3}$ (2) $\frac{1}{3}$ (3) $\frac{5}{3}$ (4) $\frac{8}{3}$

20. If the function $f : [1, \infty) \rightarrow [1, \infty)$ is defined by $f(x) = 3^{x(x-1)}$; then $f^{-1}(x)$ is :

- (1) $\left(\frac{1}{3}\right)^{x(x-1)}$ (2) $\frac{1}{2}\left(1 - \sqrt{1 + 4\log_3 x}\right)$
 (3) $\frac{1}{2}\left(1 + \sqrt{1 + 4\log_3 x}\right)$ (4) not defined

SECTION-II

(Integer Type Questions)

This section contains **Five (5)** questions. The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto one digit

21. A, B, C are sets of letters needed to spell the words STUDENT, PROGRESS and CONGRUENT respectively. Then $n(A \cup B \cup C) =$
22. A relation on the set $A = \{x : |x| < 3, x \in \mathbb{Z}\}$, where $\mathbb{Z}$ is the set of integers defined by $R = \{(x, y) : y = |x|, x \neq -1\}$. Then the number of elements in the power set of R is :
23. Consider functions $f_1(x) = \frac{1 - \sin x}{1 + \sin x}$, $f_2(x) = \frac{1 - x^2}{2 + x^2}$, $f_3(x) = 1 - x$, $f_4(x) = \frac{1}{2} \ln \left(\frac{e^x + e^{-x}}{e^x - e^{-x}} \right)$. How many of these functions are inverse of itself ?
24. If $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 + 3$ and $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = 2x + 1$ then $f^{-1} \circ g^{-1}(23)$ is equal to
25. Let f be a function defined from $\mathbb{R}^+$ to $\mathbb{R}^+$ and $(f(xy))^2 = x(f(y))^2 \forall x, y \in \mathbb{R}^+$. If $f(2) = 6$ then $f(50) =$

Answer Key DPP_03

1.	(1)	2.	(3)	3.	(3)	4.	(4)	5.	(2)	6.	(4)	7.	(3)
8.	(1)	9.	(1)	10.	(3)	11.	(3)	12.	(1)	13.	(3)	14.	(2)
15.	(1)	16.	(1)	17.	(3)	18.	(1)	19.	(3)	20.	(3)	21.	(11)
22.	(16)	23.	(2)	24.	(2)	25.	(30)						